

EYE CONTROLLED WHEELCHAIR USING RASPBERRY PI

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Abstract—The purpose of this eye-controlled wheelchair is to eliminate the need for support for people with disabilities. The proposed wheelchair control technique depends on eye movements. The camera is mounted on the user's head to take eye photos and track eye pupil movements using face landmarks recognition technology. Based on eye pupil movements, the wheelchair motor moves left, right and forward. For security, an ultrasonic sensor is introduced at the front of the wheelchair to identify obstruction and consequently stop the wheelchair. The complete system is controlled by Raspberry Pi.

Index -Dlib library, Numpy library, Open Computer Vision Library, Python, Raspberry Pi, Wheelchair.

I.INTRODUCTION

Wheelchairs controlled by eye movements allow people who are totally paralyzed to make their lives more independent and easily accessible [1]. Someone who cannot walk and cannot use a wheelchair exert a lot of physical strength to turn the wheel [14]. Disabled people will save a lot of energy.

At present there are several other methods for tracking eye pupils [3] [4] that are used to control wheelchairs, such as the eyeball method based on EOG, ECG and EEG [5] [6], decisions about the position of the pupils of the eye depend on changes in voltage. However, different output voltages are generated for different users, which leads to the wrong eye pupil position [7]. The head movement system has restrictions when the user cannot physically access the system [8] [3]. A voice-activated wheelchair that functions correctly when the user says commands clearly, moves left, right and forward [13]. Other sounds from the area can interfere with the system. Infrared reflection, which is based on eye pupil detection, tracks the exact position of the pupil of the eye. However, infrared reflection can affect the eye and that person can lose eye visibility [13].

The camera captures real-time video images such as face, eye and eye movements with a minimum delay [13] and analyzes and processes the image to send commands to the GPIO pin and then to the motor driver IC to perform various operations such as forward, left, right and stop. This system includes multi-stage who follow the center of the pupil of the eye [9]. The OpenCV library is used to recognize faces and eyes [2]. Various applications and algorithms are used to determine the exact location of eyes. Face landmarks are used that can describe facial features.

This program uses facial training set to determine where certain points of facial structure exist [12]. Figure (1) shows a system architecture diagram.

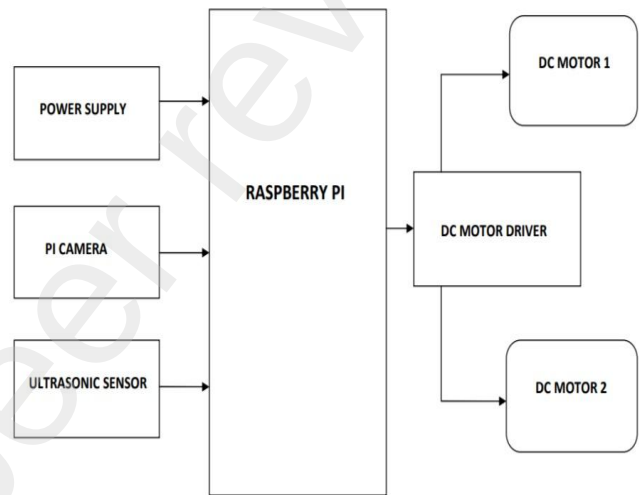


Fig.1. System Architecture diagram

The Raspberry Pi Board is the heart of the system, which controls the entire work of the system. The camera takes the picture. The various operations are performed on this image to get the output signal. The Raspberry Pi then analyzes the output signal and sends the control signal to the motor drive circuit based on the eyes position. It responds to motor movements in a clockwise or counterclockwise direction or stops. Two separate motors are mounted on each wheel. Ultrasonic sensors are also installed in a wheelchair to detect obstacles. When the sensor detects a barrier near the wheelchair, it gives a Raspberry Pi signal and sends a command to the motor driving circuit to stop the motor.

II.SYSTEM DESIGN MODEL

The system is fully independent and all modules work independently of each other. In this system, the power supply of each component is mandatory and standard power must be used for Raspberry Pi, Pi camera, sensors, and motors. The figure (2) shows working of the system.

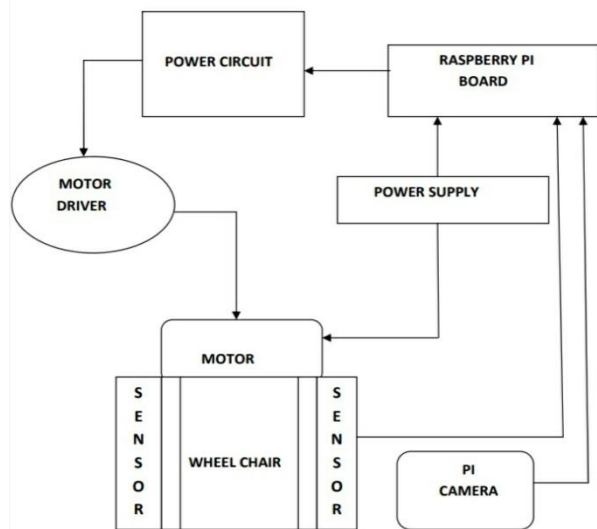


Fig. 2. Design Model

1) HARDWARE DESCRIPTION

1. Raspberry pi Board:

The Raspberry Pi card considered as the heart of the system is a single board computer that runs under the LINUX operating system. The Raspberry Pi controls the motor driver circuit which activates the GPIO pin of the Raspberry Pi. The Raspberry Pi B model only processes frame by frame.

2. Pi Camera:

The Pi camera module is a lightweight portable camera that supports Raspberry Pi. It interacts with Raspberry Pi via the MIPI camera serial protocol. Usually used in project supervision.

3. Ultrasonic Sensor:

Ultrasonic sensors detect obstacles in the way of the wheelchair system. The ultrasonic sensor is connected directly to the Raspberry Pi. It Receives information and measures the distance between a wheelchair and obstacles. If obstacles are detected near the wheelchair, the motor stops driving the wheels.

4. DC Motor:

Two 12 V DC motors show wheelchair movements in front, back, left and right. The L298N motor driver is used to interact with the Raspberry Pi board. L298N is a dual H-bridge driver that supports speed and direction control of two DC motors simultaneously. This module can drive DC motors with voltages in the range of 5 to 35 V with peak currents of up to 2A.

2) SOFTWARE DESCRIPTION

1. Putty Software:

Putty is a terminal emulator and free open source application for transferring network files. The Putty software is used to connect the desktop to the Raspberry Pi.

2. OpenCV Image library:

OpenCV is a Python development project library to fix computer vision problems. It has C, C++, Python and Java interfaces and supports Windows, Linux, Mac OS, iOS and Android. The Python Opencv 3.4.3 library is used on this system.

3. Python Language:

Python is an object-oriented translator, a high-level programming language with dynamic semantics. High-level information structure, combined with dynamic recognition and dynamic construction, makes it very attractive for fast applications. Python is easy to use and useful for debugging.

4. Numpy library:

Numpy is a Python tool which means "Numerical Python". This is the main library of scientific calculations that contains powerful objects from n-dimensional arrays and provides tools to integrate C, C++, and more. This is also valuable in linear algebra, any possible numbers, etc.

III. DESIGN METHOD

The principle of this system is eye tracking, movement, blinking of the eyes, and especially eye tracking based on computer vision technology. A new algorithm used for recognizing the position of the eye by facial landmarks. Various steps for detecting eye movements, such as face and eye detection, gray and threshold conversion of eye. The first system recognizes images captured using a Raspberry Pi camera. The first goal is to find the face, and then determine the correct position of the eyes. Find facial landmarks after importing opencv, numpy library and dlib library. The camera takes an image. Facial landmarks are used for face and eye recognition. For that face detector and landmarks predictor used to find the landmarks. The system takes real time frames from a Raspberry Pi camera. Then the frame is converted to a gray frame. Because it is lighter than color image and saves processing power. Using face landmarks recognition technology, the system can identify 68 specific face landmarks. Each point has a specific index.

Many software libraries can display important features of a face in an area of interest. The Python dlib library implements this function using Kazemi and Sullivan's face alignment in one millisecond with a group of regression trees [12]. The program uses facial training set to determine where certain points are in the face structure. Then the program draws the same points in the area of interest of another image. The program uses priors to estimate the possible distance between key points [10]. The library displays a 68-point plot from the input image.

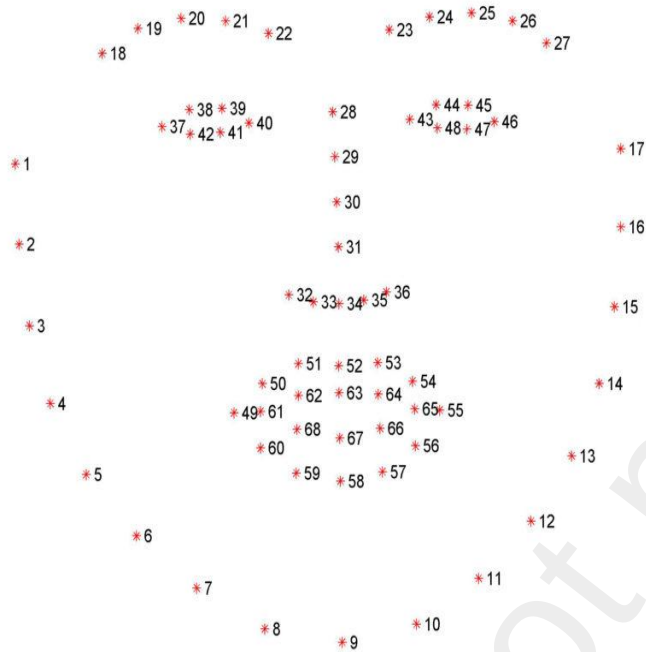


Fig.3. Dlib Facial Landmarks Plot

1) Detecting Eye Blinking with Facial Landmarks

Eye blinking can be detected by looking at important facial features. For eye blinks we need to consider points 37-46, which describes the eyes. If the eyes blink in real time using facial landmarks [11], Soukupová and Čech adopt an equation that represents the aspect ratio of the eyes. Eye Aspect Ratio is an assessment of the condition of opening eyes. As shown in figure (6), the eye aspect ratio can be determined from the following equation.

$$EAR = \frac{\|p_2 - p_6\| + \|p_3 - p_5\|}{2\|p_1 - p_4\|}$$

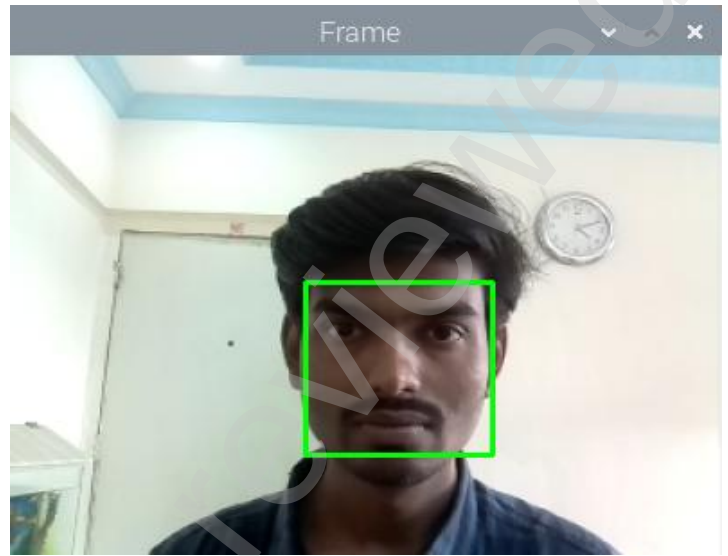


Fig.4. Face Detection

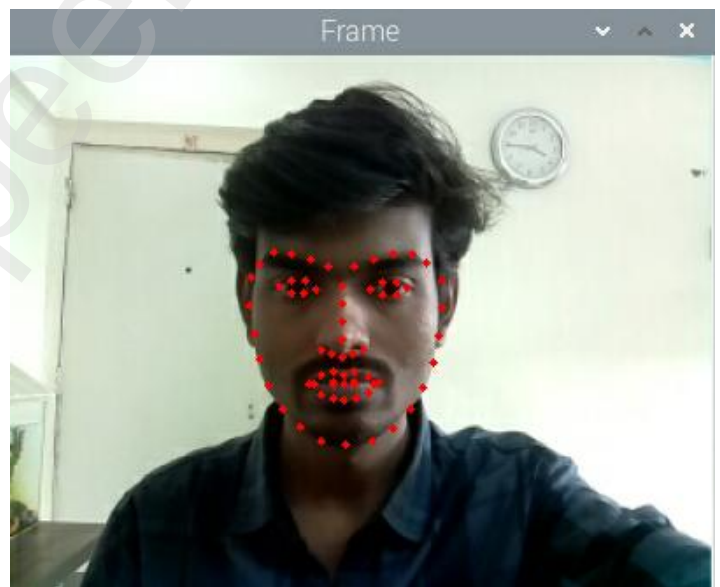


Fig.5. 68 Facial Landmarks Points Plot

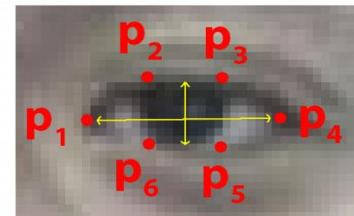


Fig.6. Eye landmarks

The eye aspect ratio is a constant value when the eyes are open, but quickly drops when the eyes are closed. Human eyes are clean. The program can determine if a person's eyes are closed when the aspect ratio of the eye falls below a certain threshold.

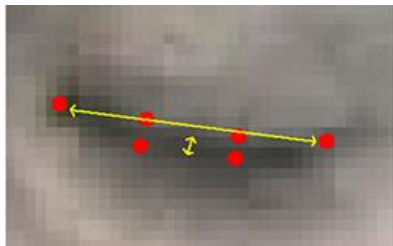


Fig.7. Eye closed

2) Detecting Eye Movement with Facial Landmarks

The region of the left eye corresponds to landmarks with indices 36, 37, 38, 39, 40 and 41. The region of the right eye corresponds to landmarks with indices 42, 43, 44, 45, 46, 47. Work in the region of the left eye. The same procedure follows for the area of the right eye. Having received the coordinates of the left eye, you can create a mask that accurately extracts the inside of the left eye and excludes all inclusions, as shown in Figure (9). Now remove the eyes from the face. You can cut a square shape in the image, so use the end points of all eyes (top left and bottom right) to get a rectangle. We also get a threshold to detect our eyes.

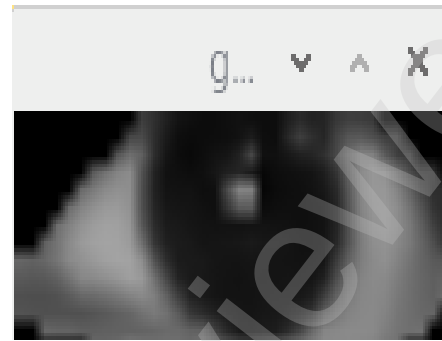


Fig.9. Gray image of Eye



Fig.10. Threshold image of Eye



Fig.8. Eye Region on Face

A) Eye Detection



Fig.11

If the sclera is more visible on the left, the eyes look to the right, as shown in figure (11). To determine whether the sclera's eyes are converted to gray levels, shown in Figure (9), a threshold is applied and white pixels are calculated as shown in Figure (10). Then separate the left and right white pixels and get the eye ratio.

B) Detecting gaze

The gaze ratio shows us where a particular eye is looking. Usually both eyes look in the same direction. Therefore, if you truly recognize the gaze of one eye, you can find the gaze of both eyes. Only when you want to be more precise you can find the gaze of both eyes and use both values to determine the gaze ratio.



Fig.12. Eye is looking at Left



Fig.13. Eye is looking at centre

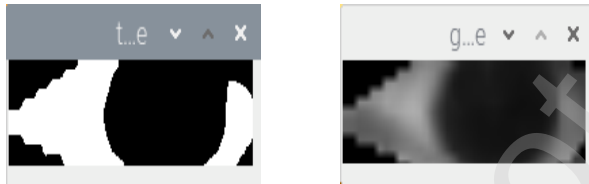


Fig.14. Eye is looking at Right

C) Eye Tracking

If the user only looks to the right on one side, the user points the pupils and iris in the right direction, while the rest of the eye is completely white. The others are irises and pupils, so it's not completely white and in this case it's very easy to find how it looks.

If the user sees from the other side, the opposite is true. When the user looks in the middle, the white portion between left and right is balanced. We take the eye area and make a threshold to separate the iris and pupils from the whites of the eyes. In this way we can follow eye movements. Based on this, the system provides instructions for wheelchair movement. Figure (12) shows the recognizable eye in the left position, Figure (13) shows the position of the pupils in the middle of the eye and Figure (14) occupies the eye position on the right side of the eye. The system works according to the eye pupil position and moves the wheelchair left, right and forward.

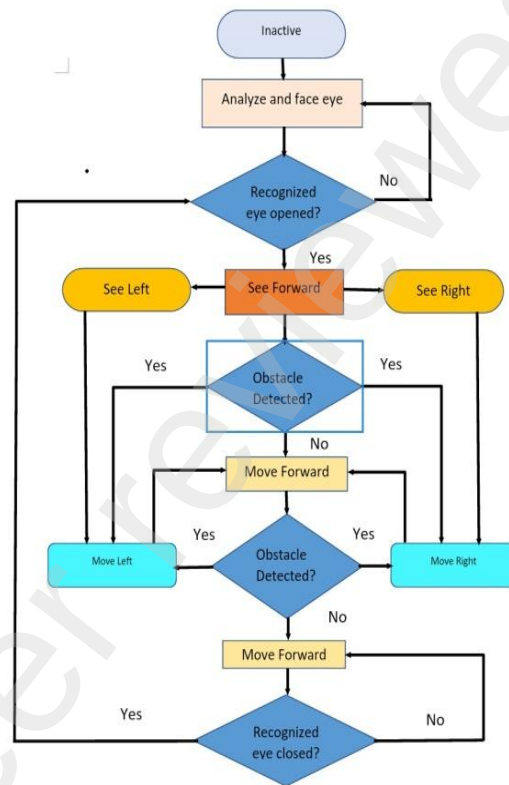


Fig.15. Flowchart of system working

When the pupil of the eye moves to the left, the wheelchair motor runs on the left side, and when the eye moves to the right, the right side of the motor must move [13]. If the eye is in the middle, the motor must also move forward. If problems are found, the system stops working. Eyes blink logic applies to starting and stopping wheelchair systems [14]. Figure (15) shows a block diagram of a wheelchair system. The raspberry pi camera is installed in front of the user. The distance between the eyes and the camera is fixed and must be in the range of 15 to 20 cm.

IV. SYSTEM INSTALLATION

To boot a raspbian image file in micro SD card win32 diskimager software is used. Then insert the micro SD card into the raspberry board and leave the Raspbian operating system directly without resetting [13].

V. RESULT

The wheelchair system accepts video processing results that are generated based on the location of the eye pupil. Thus, sending orders to the motor driving circuit for the wheelchair movement. An exact illustration of the results of video processing is shown in Figure (16).



Fig.16. Output of the system

Ultrasonic sensors detect obstacles. And it measures the distance between a wheelchair and an obstacle. If the obstacle is extremely near to the wheelchair, the motor stops the wheelchair.

VI.CONCLUSION

In this system, we have developed a wheelchair system that allows disabled and physically handicapped people to move their wheelchairs in the desired direction. Places in the dark light interfere with the operation of the wheelchair. This system works perfectly with ambient light and indoors.

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